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Towards the National Higher Education Database in Indonesia: Challenges to Data Governance Implementation from The Perspective of a Public University

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Abstract

Higher education institutions in Indonesia are required to report their academic data to the national higher institution database (namely PDDikti) through a synchronization mechanism. Because the PDDikti database has been used as the reference for nationwide programs such as university ranking, diploma verification, faculty promotion, and tenure, academic data in their reports have to be valid, accurate, and up to date. Yet, data quality is an issue among higher institutions, while data governance is the heart of data quality. This paper investigates challenges encountered by a public university in Indonesia in implementing data governance and how to resolve them. The findings outline six main challenges in data governance implementation.

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1. Introduction

Through the regulation on National Higher Education Database [1], the Directorate General of Higher Education (DGHE) aimed to collect higher education-related data from all entities under their hierarchical direction, including all public and private universities in Indonesia. This regulation instigated a significant implication for all higher education institutions, i.e., they must regularly report their academic-related data to the central government. The

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reporting process is conducted automatically through a synchronization mechanism using a feeder called PDDikti feeder [2]. In order to preserve its quality, the reported data must be complete, valid, accurate, and up-to-date [3].

However, synchronizing data across government sectors at various levels is a challenging task, including file synchronization and sharing practices [4]. Some challenges impede the implementation, for example, local governments experienced technological challenges, such as data incompatibility, low data quality, and ill-defined data management [5], and also lack of technical capacity and open data plans [6]. In the context of synchronizing higher education-related data, the synchronization process between higher education's academic systems at the university level and the central government's national HE database at the national level was also problematic, such as lack of technical skills [7], and data migration issues [8,9].

These problems indicate the importance of data governance at the higher institution level. Data governance combines policies, procedures, rules, regulations, standards, and other arrangements to manage data and ensure data quality, for example, as depicted in the case study of data governance in the educational sector [10]. In addition, data governance has raised its importance in the cloud computing era [11]. However, the implementation of data governance is still found challenging, from having a clear understanding of the terminology of data governance [12] to implementing activities related to data governance [13]. Understanding the barriers to data governance practice is essential because the successful implementation of the national higher education database depends on the supply of data from universities [14]. Nevertheless, to our best knowledge, no literature discusses the barriers to data governance implementation in higher education institutions in Indonesia.

This paper seeks to answer the following research question: "What are the challenges experienced by public universities in their efforts to implement good data governance?". To answer the question, this study employs a case study, that is, a public technical university located in the eastern region of Indonesia. This study is interested in discovering the experience of the technical university in dealing with the implementation of data governance to respond to the whole-of-government data integration imposed by the central government.

2. Data Governance Literature

This section outlines the case study on nationwide program of governing higher education data from the perspective of Directorate General of Higher Education (DGHE), followed by describing terminology related to data governance from previous literature.

2.1. National Higher Education Database in Indonesia (PDDikti)

A national higher education database is the first stage to achieve One Data in Higher Education, while there has been discussions on big data and data analytics in higher education institutions [15]. There are several goals that the Directorate General of Higher Education (DGHE) wants to accomplish by having the national database, two of which are: 1) to efficiently control higher education institutions, 2) to facilitate data sharing across government sectors at all levels, and, therefore, to increase data quality [16]. To create the national database, the DGHE requires valid, accurate, and the most recent data from all higher education institutions in Indonesia. They are required to regularly report data to the DGHE's platform called PDDikti (the portal of the national database of higher education) through a synchronization mechanism. Data to be reported to PDDikti comprises academic data, teaching, and learning facilities, students' profiles, and organizations-related data such as study programs, colleges, schools, etc.

2.2. Data Governance

Data governance is the governance of technology and people [17]. It specifies organizations' accountabilities and decision rights regarding data [18]. In public services, data governance has emerged because of several reasons, such as the organization's growth, regulatory compliance, and integration with other cross-functional systems [19]. There are various components of data governance. It is the combination of policies, procedures, rules, regulations, standards, and other arrangements in managing data [17]. Another study also contributed to understanding data governance, for instance, it is termed as defined rules, processes, accountabilities, decision rights, monitoring, controlling or other enforcement methods, and organizational bodies [20]. To sum up, the data governance model or framework consists of three essential components, i.e., policy, processes, roles, and responsibilities [21].

Data quality is the heart of data governance: “managing the quality of an organization’s data is not operational without a formal data governance model” (p. 6) [20]. A previous study also has proposed six attributes of good data quality, i.e., accuracy, timeliness, relevance, completeness, understood, and validity [3]. However, several data quality problems and barriers could impede data governance implementation [22]. It is also important to balance the risks and opportunities of implementing data governance programs [23].

A previous study has defined ten essential success factors of data governance, i.e., accountability and strategic accountability, standards, managerial blind spot, embracing complexity, cross-divisional issue, metrics, partnership, choosing strategic points of control, compliance monitoring, and training and awareness [24]. In addition, a study has summarized the work of previous scholars in 14 CSFs to implement data governance in an organization [20]. This paper explores phenomena in the case study based on the concepts of data governance components, data quality barriers, change management issues, and CSFs. The summary of the required concepts taken from the literature is presented in Table 1. This table serves as a guideline for the interview process and data analysis.

Table 1 The determinants of good data governance [3,20,22,23]

Category	Determinants
Organization	(1) Roles & responsibilities, (2) Business and IT involvement, (3) Cross-business units coordination, (4) Change management plan, (5) Resistance to change, (6) Communication plan, (7) Company culture (in data quality), (8) Training, (9) Metrics and measures, (10) Priority issue (working on silos), and (11) Organizational structure
Resources	(1) Funding, (2) Skills, Knowledge, and Experience, and (3) People
Top Management	(1) Executive support and leadership, (2) Mission, vision, and (3) Commitment and Priority
Policy, Regulation, Business Process	(1) Data governance understanding, (2) Policies, procedures, process, and (3) Compliance enforcement, 4) Incentives, 5) Standards and guidelines, and 6) Clarity and understanding of business processes
Technology and Data	(1) Quality control, (2) Internal data source, (3) Data quality measurements, (4) IT systems for data management, (5) Legacy systems, (6) Data dictionary, metadata, (7) Data quality (8) System integration
External	(1) Intermediate application issue, (2) Operational issues, (3) Metrics and measures, (4) Data dictionary, metadata, and (5) Organizational structure

3. Methods

The research question of this study seeks to discover the experiences of a public university in dealing with the implementation of data governance to respond to the whole-of-government data integration imposed by the central government. For this purpose, qualitative research using a case study is used because a case study is suitable for exploring phenomena in a real-world context [25,26]. This study is a single case study, so the organization being observed is a single organization. This paper presents an exploratory study to investigate issues and practices of data governance implementation at the university level. For this purpose, qualitative research using a case study is used because a case study is suitable for exploring phenomena in a real-world context [25,26]. This study is a single case study, so the organization being observed is a single organization.

The case study is a technical and public university in Indonesia with a direct coordination line with the DGHE. Since 2013, they have started integrating their systems and data, shifting from silos to centralized systems. The academic system of this university has been perceived as mature, with the system being adopted by several universities in Indonesia. However, even though this university is leading in the implementation of IS/IT for higher institutions, the policy imposed by the DGHE related to synchronization and integration with PDDikti implies a lot. One of them is that the university started to realize the importance of data governance, and therefore, in 2017, they kicked off the plan and implementation and currently, are continuing the efforts to improve their data governance.

The data used in this study is collected from two sources, i.e., (1) the existing documents and (2) semi-structured interviews with data management officers in the case study. The relevant documents include a research report on data integration and reporting and a draft of data management guidelines. Both documents are very technical, so interviews are conducted as a counterpart. In this study, the interviewee is the Head of Big Data Management. This person has

the authority and responsibility to manage, implement data governance at the institutional level and communicate with the central government. The interviews were done online through Zoom three times, recorded in notes and a digital recording, with consent from both the interviewer and interviewee.

The study uses a content analysis [27], classifying many words into a few content categories [28]. The next stage is to determine the coding rules based on the category to achieve consistency and avoid bias in the analysis. The coding rules are constructed based on synonyms and words with similar contexts. Some words in the coding rules were also derived from the interview results mentioned by the interviewee.

4. Results and Analysis

This section outlines results from the study by pointing out six challenges in implementing data governance. In each challenge, qualitative analysis is provided by outlining findings from data collection.

4.1. Organizational-related Challenges

It was noted from the interviews that data quality has always been the major concern of PDDikti. In creating high data quality on the PDDikti system, cooperation between DGHE and higher institutions is critical. However, many cases occurred when universities relied on one unit, mainly the Information, Communication, Technology & Services (ICTS) unit. PDDikti was not perceived as the university agenda. The business and IT people must be on the same page. The action taken by the university was to bring the issues to the executive leaders.

“The ICTS could not do this alone. Our tactic was to bring the issue of PDDikti to the executive leaders whenever I had meetings with them. Initially, our team approached the Vice Rector of HR, Organization, and Information Systems/Technology to gain support. Next, we approached the executive leaders in academic affairs, the deans, the head of departments, and the program directors. The best way to convince them is to visualize the positive and negative impacts if we do not take the action seriously. We were lucky because there was a window opportunity at that time, our university was busy with achieving the best university ranking. PDDikti is one of Indonesia's most important sources for university ranking.” (Interviewee)

It is also indicated from the interviewee's statement, noting the impact of organizational restructuring and the clarity of roles and responsibilities in reporting data to PDDikti. It was also pointed out that, at first, there was weak cross-unit coordination which emphasized silos systems. The university defined the data governance steering committee to address this issue, including data stewardship. Data stewardship determines data management's formal roles and accountability in the case study. There are three types of data stewards, i.e., coordinating data stewards, business data stewards, and technical data stewards. Table 2 presents the steering committee of data governance.

Table 2 Data governance steering committee

Roles	Description	Example (Unit)
Data Owner	Those who have the rights to make decisions regarding rights to access and modify data and how the data is used.	Chief Executive of Universities (In Indonesia: Rectorate)
Coordinating Data Steward	A unit responsible for coordinating the activities of managing academic data.	Academic Unit
Business Data Steward	A unit that manages data transactions so that the business can function	Department's Academic and Student Affairs
Technical Data Steward	A team/unit accountable for technical systems	Information, Community, Technology, and Services (ICTS)
Data Supplier	The providers of data	Students, Lectures, Departments

In the case study, the data stewardship's roles are represented formally in a structural position. It means they will receive incentives based on their position. This approach was taken to follow up on resistance that occurred among units. At first, some did not want to receive more responsibilities regarding the implementation, as it would add to

their workload. However, assigning someone with a formal position has resolved the issue. This arrangement required the involvement of executive leaders. The change management plan of data governance implementation was initially known by the ICTS unit. Communicating it with all stakeholders has facilitated a better understanding of the organizational transformation towards good data governance.

Initially, the communication was done haphazardly due to no clear guidance regarding the roles and structure of stakeholders involved in data governance. Later, as the data governance steering committee of the university was defined, the university also formed the structure of the steering committee, as depicted in Fig. 1. There is also a communication mechanism using social media groups. Each stakeholder in data stewardship needs to assign a representative in the group.

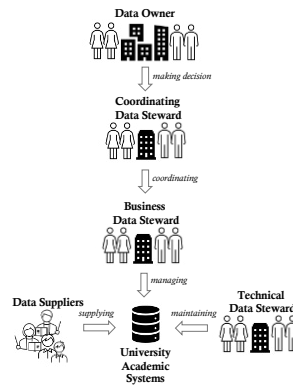


Fig. 1. The structure of the data governance steering committee

Another issue at the organizational level is the lack of data-driven culture, especially from the data suppliers' side. The interviewee indicated that students' awareness regarding the correctness and completeness of their data was low. In Indonesia, some jobs require validating the candidates' diplomas through PDDikti. This was problematic when students were too late to realize their data on PDDikti was inaccurate. As the change request of modifying PDDikti's data takes time, it might cause negative consequences for students, as indicated by the interviewee below.

"Our students' awareness regarding their data is still low. They noticed the inaccuracy of their data when they were about to graduate. The process of changing data on PDDikti always takes time. Yet, as the complaint was reported right before graduation, it was hard to change PDDikti's data quickly. As a result, some of them graduated without the diploma listed on the PDDikti yet." (Interviewee)

Training and workshops are essential in data governance implementation as educating people who are in charge of data governance will get used to the viewpoint of a data-driven culture. In addition, training is a means to facilitate data stewards to comprehend the changes, the expectations towards them, and how the changes impact them. Regarding training, the interviewee remarked on variability in IT skills among data stewards. There was a significant skill gap even between those with the same position, e.g., operator data at the department level. In addition, the training specific for data synchronization with PDDikti was intended for IT people but not for non-IT people involved in data governance. The university conducted IT training at different mastery levels to overcome the barriers. The training was performed in collaboration between the HR Unit and the ICTS. As a unit that knew the competence of staff in the university, HR provided the selection and classification according to their competence. The ICTS, on the other hand, served as the trainer of the training. Some training that had been conducted included basic computer training, intermediate level computer training, and workshops on PDDikti for business data stewards, etcetera.

4.2. Resources-related Challenges

The process of integrating university data and PDDikti is a complex task. The technical work is arduous, especially when the university has a lot of information systems that have not been fully integrated. In addition, from around 2017 to 2024, the focus of the extensive data technology section of the ICTS unit is to integrate data from various

information systems, ranging from academic, finance, human resources, research, and other services. Furthermore, ranking systems from multiple agencies, such as the DGHE's ranking systems and accreditations of universities and departments, have occupied most of the working time. Unfortunately, the number of staff was too limited compared to the heavy load this section had to carry. In order to address staffing shortages, the ICTS unit proposed additional staff to the executive leaders. The interviewee mentioned that the process was not easy. Therefore, we also relied on students by opening internship positions. In addition, the ICTS unit also received a budget specific for PDDikti-related programs. The university allocated the funding to the division of big data management of the ICTS unit.

"Each December, we propose a budget to the executive leader and target it to develop tools, capacity building, and other PDDikti-related programs. If the budget is tight, we prioritize purchasing a license for the intermediate application that bridges our systems and PDDikti." (Interviewee)

Additionally, there was also an issue regarding skills and knowledge. Digital literacy and skills varied, especially those of business data stewards. Moreover, several of them were not familiar with working with computers. As a result, efforts had to be made through training and workshops to reduce the gaps in skills. Workshops and meetings in the form of focus group discussions were also conducted to increase awareness and responsibilities regarding their roles as data stewards. For example, workshops were aimed at informing them how to ensure the correctness of data, report data to PDDikti, etcetera.

4.3. Top Management-related Challenges

On top of everything, executive management's support is the most essential aspect of successfully implementing data governance. The interviewee indicated that, at first, the top leaders paid less attention to the importance of PDDikti. Then, there was a transferred authority in the university where the authority holder shifted from the ICTS Unit to the academic unit. The one who reported the data was also directed to the department, no longer the ICTS Unit. Involving important stakeholders in data stewardship was also considered a successful approach as it could increase the sense of belonging that paying attention to PDDikti is the responsibility across units in the university. The leaders also agreed to release a formal document determining the roles and responsibilities of those involved in data governance implementation.

"Management support is everything. However, we must know which stakeholders have the highest interests, power, and support. Using a soft approach to transfer the authority. We positioned the academic unit as the one that held the authority to deal with PDDikti. This approach made the academic unit feel important and bravely take more progressive stages." (Interviewee)

In addition, linking the vision and mission of data governance with the mission and vision of the university in achieving the best rank in any ranking system was perceived as important. The interviewee indicated that the university data on PDDikti was the determinant of university ranking.

4.4. Policy, Regulation, and Business Processes-related Challenges

Data governance is an unfamiliar term for many people, especially non-IT people. The interviewee stated that the understanding of stakeholders regarding data quality is very low. Many cases were found where data was invalid, redundant, missing, etcetera. Data suppliers also sometimes did not care whether their data was correct. To increase the understanding of data governance, the interviewee targeted to increase the awareness of the executive leaders first. Whenever they received support, the data governance team found it easier to promote some changes required to achieve better data quality to operators in departments, units, or other data stewards. Some changes, including adjusting existing business processes, needed to be done. The business processes could be easily adjusted whenever the roles and responsibilities of data stewards have been formally determined.

Discussing policies, procedures, and standards, the interviewee mentioned that these instruments are essential in implementing good governance. He noted some problems had occurred due to the absence of policies and procedures, such as an ambiguity in roles and responsibilities where people refused to do some tasks due to no clear or legal basis. Another problem is ambiguity in managing data. For example, it was not clear who managed the master data of academic affairs and who could have access to the database. At that time, there were duplicate roles between units. The solution is to have a formal data governance document and publish it. Regarding an enforcement mechanism, the

interviewee indicated a mechanism where students are forced to check their PDDikti data at the beginning of the semester through a form that has to be answered and validated by the students.

4.5. Technology and Data-related Challenges

The interviewee noted that many data sources for PDDikti reporting were initially invalid and incomplete. The causes were many, for example, no clear matrix or measurement for data quality, and low-quality assurance as the job related to PDDikti reporting was only done by a few people who knew nothing about the validity of the data. Another reason was redundancy in databases, for example, academic databases, where there were more than technical data stewards at that time. Somehow, the data used for PDDikti reporting is also related to data outside the academic systems. However, challenges occur whenever the systems and databases are not integrated. In the case study, it was noted by the interviewee that the database in legacy systems was not integrated and that influenced the quality of the data itself.

4.6. External Challenges

The data reporting to the DGHE depends on the quality of the systems and services of PDDikti too. The challenge could be, for example, the feeder was not reliable, or changes frequently occurred. The interviewee stated that his university relied on an intermediate application. Thus, the changes and problems on the PDDikti side impacted the intermediate system. Another challenge was the lack of responsiveness to PDDikti's helpdesk system. Whenever there was an issue, the resolution time could be so long. Next, sometimes there were times when targets from central governments or other external entities came simultaneously. The interviewee perceives good data governance to be helpful in dealing with sudden requests from others. This is because the responsibility of data has been divided and therefore, does not lean on one unit, the ICTS Unit.

Appendix A presents all challenges and solutions from six perspectives, including a) organization, b) resources, c) top management, d) policy, regulation, and business processes, e) technology and data, and f) external challenges.

5. Conclusion

This paper answers this research question: "What are the challenges experienced by public universities in their efforts to implement good data governance?". Referring to this question, two points are discussed in this paper. Using qualitative research, this paper discusses challenges in implementing data governance from the perspective of a public university. The results indicate that implementing data governance is not only about technical arrangements and their issues. More than that, the challenges from the organization's side are various. Most challenges are related to the organization's issues, such as no clear roles and responsibilities, business and IT were not on the same page, no awareness regarding data governance change, resistance to change, etcetera. Other challenges are technical, policies and regulations, top management, resources, and external issues. This paper also identifies the solutions the case study took to implement data governance. This study is an early step in identifying challenges in implementing data governance, which can contribute to better understanding the role of data governance in data science practice [29]. In terms of academic contribution, currently, most literature in data governance lists success factors of the implementation. This paper contributes to enrich academic literature in data governance, especially with regards to challenges in implementing data governance in higher education institutions.

This paper relied on a single case study, and the sources of data collection were documents and interviews. However, relying on a single case study has a limitation in terms of the ability to generalize findings. Future work can be done by adding more other universities as case studies. This way, generalization of the challenges in implementing data governance can be obtained. In addition, it is also possible to create a framework of data governance implementation in higher institutions, mainly to deal with the synchronization and integration with the central government's data, aiming to improve the quality of education data [21]. With the advance of cloud computing, it is also worth to discuss cloud data governance [11], and thus aiming to identify to what extent data governance practices implemented in higher education institutions.

Appendix A. Challenges and Solutions in Implementing Data Governance

Theme	Determinants	Barriers	Solutions
Organization	Roles & responsibilities	No clear roles and responsibilities	(1) Bringing the issues to the executive leaders, finding the most appropriate windows of opportunity, (2) Determining data stewardship and data steering committee, including their roles and responsibilities, (3) Determining the structure of data governance's steering committee, (4) Quality control is directed to business data stewards, (5) The responsibility to report/synchronize the data is directed to the department's program study (business data stewards), (6) Data steward's roles are represented formally in a structural position, supported by legal base and incentives, (7) Data governance group in which there is a representative of each data steward, (8) Conducting training, workshops, meetings, (9) Collaboration with other units (e.g. HR) in competency building
	Business and IT involvement	Business and IT were not on the same page	
	Cross-units coordination	Silos, only ICT unit or academic unit that holds the responsibility	
	Priority issue (working on silos)	Management did not see the data governance issue as a priority issue, the responsibility is given to specific unit	
	Organizational structure	No clear structure on data governance	
	Communication plan	No communication structure	
	Resistance to change	Other units are resistant to receive more responsibilities regarding the implementation	
	Change management plan	No understanding regarding data governance change	
	Company culture (data quality)	The culture of assuring data quality is not yet formed, some units do not want to integrate their data	
Resources	Training	There is a training for ICT unit, but no training for units who own the data	
	Funding	Limited funding	(1) Prioritizing budget, for example, the budget is prioritized to purchase annual license for the intermediate application, (2) Training and workshop, (3) Adding non-permanent staff, interns.
	Skills and Experience	Less skills related to assuring data quality	
	People	Too many works that hinder the responsible people to focus on data governance implementation	
Top Management	Executive Supports	Lack of support from top management	(1) Building sense of importance to important stakeholders and sounding to them as many times as possible, (2) Transferring authority by emphasizing that data validity is the responsibility of business data stewards, (3) Linking data governance's mission and vision to university ranking.
	Mission, vision	No clear mission and vision	(1) Gaining awareness of the leaders first, then staff at the lower level, (2) Having a formal data governance document or start it with
	Organization's Priority	Department or unit's lack of priority	
Policy, Regulation,	Data governance understanding	Units do not understand data governance and why it is importance	

Theme	Determinants	Barriers	Solutions
Business Processes	Policies, procedures, process, standards	No instruments or if there is, not yet formalized	regulation (Indonesia: Surat Keputusan), (3) Integrating enforcement mechanism with academic information systems, (4) Provide a legal basis (Surat Keputusan) regarding incentives, (5) Together with relevant units, determining business processes
	Compliance enforcement	Enforcement is hard to implement	
	Incentives for ensuring data quality	No incentives	
Technology	Business processes	Internal business processes are unclear, incomplete, wrong; Overlapped business processes	
	Quality control	Lack of quality control (Only few people who are accounted for it)	(1) Letting data suppliers and business data stewards ensure the validity of their data, creating enforcement in checking data quality regularly, could be integrated with academic systems, (2) A policy that regulates centralization of information systems and databases in the university, (3) Strengthening coordination across units, (4)
	Internal data source	Redundant, inconsistent data in many internal databases	Clarifying and creating a data dictionary document that has been discussed with relevant units, (5) Formalize the business processes of PDDikti reporting, including the data stewardship, (6) Assigning business data stewards as the quality control of PDDikti-related data, (7) The need to have additional system to bridge between the intermediary system of central government with the internal system.
	Data quality measurements	No clear measurement for data quality	
	IT systems for data management	Existing information system cannot directly link to feeder	
	Legacy systems issue	The databases in legacy systems are not integrated	
External	Data dictionary, metadata, data agreement	Unclear data definition that results in data redundancy	
	Data quality issues	Data is not valid, while the ICT unit does not know about the validity of data	
	System integration issue	Existing information systems are not yet integrated	
	PDDikti application issue	Bugs, errors on PDDikti systems due to frequent changes	(1) Contacting vendor that provides the intermediate application to fix the issue, (2) Having contacts of PDDikti's personnel, asking persons with authority to convey the message to the PDDikti team
	Operational issues	An issue with the intermediate system due to frequent changes on PDDikti systems	(3) Planning ahead of time, working together with other units (tasks division).
	Metrics and measures	Lack of responsiveness from helpdesk system of PDDikti	
		Pressure from central government (deadline, KPIs)	

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